

Erdős Problem #976

EP-976

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f \in \mathbb{Z}[x]$ be an irreducible polynomial of degree $d \geq 2$. Let $F_f(n)$ be maximal such that there exists $1 \leq m \leq n$ with $f(m)$ is divisible by a prime $\geq F_f(n)$. Equivalently, $F_f(n)$ is the greatest prime divisor of $\prod_{1 \leq m \leq n} f(m)$. Estimate $F_f(n)$. In particular, is it true that $F_f(n) \gg n^{1+c}$ for some constant $c > 0$? Or even $\gg n^d$?

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